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Ex4b: Information Retrieval using spaCy

1. Learning Objectives

- To understand Information Retrieval (IR) using the spaCy NLP library.
- To preprocess Amazon Fine Food Reviews using tokenization, lowercasing, stopword removal, non-alphabetic filtering, and lemmatization.
- To convert cleaned reviews into TF-IDF vectors.
- To preprocess a user query using the same NLP pipeline.
- To calculate cosine similarity between the query and reviews.
- To retrieve the top-k most relevant reviews.
- To analyze the retrieved reviews along with their similarity scores and metadata.

2. Methods Used / Python Libraries Involved

Library / Method	Purpose	Reason to Use
Pandas	Load and manipulate the dataset	Makes it easy to handle CSV data and DataFrames
spaCy	Perform NLP preprocessing	Provides efficient tokenization, stopword detection, and lemmatization
token.is_alpha	Filter tokens	Removes numbers, punctuation, and special characters
token.lemma_	Perform lemmatization	Converts words into their base/root form
argsort()	Sort similarity scores	Helps identify the top-k relevant reviews

3. Program Code

```
import pandas as pd
import spacy
 from         sklearn.feature_extraction.text         import
TfidfVectorizer    from    sklearn.metrics.pairwise    import
cosine_similarity

nlp = spacy.load("en_core_web_sm")

df = pd.read_csv("Reviews.csv") df

= df.dropna(subset=["Text"]) df =

df.head(20)

print("Number of reviews:", len(df))

 def preprocess_text(text):

doc   =   nlp(text.lower())

cleaned_words = []

    for token in doc:
 if token.is_alpha and not token.is_stop:
            cleaned_words.append(token.lemma_)
 return " ".join(cleaned_words)


df["Cleaned_Text"] = df["Text"].apply(preprocess_text)


vectorizer = TfidfVectorizer()

tfidf_matrix  =  vectorizer.fit_transform(df["Cleaned_Text"])

print("TF-IDF Matrix Shape:", tfidf_matrix.shape)


 def retrieve_reviews(query,
top_k=5): cleaned_query =
preprocess_text(query)
 print("\nOriginal   Query:",   query)   print("Processed
    Query:",      cleaned_query)      query_vector      =
    vectorizer.transform([cleaned_query])
```

```
    similarity_scores = cosine_similarity(
        query_vector, tfidf_matrix
    ).flatten()
top_indices   =   similarity_scores.argsort()[-top_k:][::-1]
    print("\nTop", top_k, "Relevant Reviews:\n")

    for i in top_indices:
print("Similarity                    Score:",
        round(similarity_scores[i], 4))
print("Original Review:")
        print(df.iloc[i]["Text"]
        )
print("\nCleaned Review:")
        print(df.iloc[i]["Cleaned_Text"]
        )
if "ProductId" in df.columns:
            print("Product ID:", df.iloc[i]["ProductId"])
if "Score" in df.columns:
            print("Score:", df.iloc[i]["Score"])
if "Summary" in df.columns:
            print("Summary:", df.iloc[i]["Summary"])
print("-" * 80)


retrieve_reviews("great product with fast shipping", 5)
retrieve_reviews("disappointed", 5)
```

4. Output

```
Number of reviews: 20
TF-IDF Matrix Shape: (20, 41)

Original Query: great product with fast shipping
Processed Query: great product fast shipping

Top 5 Relevant Reviews:

Similarity Score: 0.8538
Original Review:
This is a great product and it arrived with fast shipping.

Cleaned Review:
great product arrive fast shipping
```

```
Similarity Score: 0.5296
Original Review:
The shipping was fast and the product was exactly as described.

Cleaned Review:
shipping fast product exactly describe
```

```
Similarity Score: 0.5055
Original Review:
Great product, excellent quality and very fast delivery.
```

5. Learning outcome

- Learned how to use spaCy for Natural Language Processing.
- Learned to perform tokenization and lowercasing.
- Learned to remove stopwords and non-alphabetic tokens.
- Understood lemmatization and its importance in text preprocessing.
- Learned to represent reviews using TF-IDF vectors.
- Learned to calculate cosine similarity between queries and documents.
- Successfully retrieved the top-k relevant Amazon Fine Food Reviews.
- Learned how NLP preprocessing can improve the effectiveness of an Information Retrieval system.